

431 Lab 6 Instructions

Fall 2026 - deadline in [Course Calendar](#)

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! Important

- This Lab contains 8 tasks for you to complete.
- The deadline for completing this Lab is posted in the [Course Calendar](#).

R Setup

We'll leave it to you to identify useful packages to add to the tidyverse for this Lab.

```
knitr::opts_chunk$set(comment = NA)

library(tidyverse)

theme_set(theme_bw())
```

Getting Help

You may discuss each Lab with Professor Love, the teaching assistants or your colleagues, but your answer must be prepared by **you working alone**. Don't be afraid to ask questions, using any of the methods described on [our Contact Us page](#).

Learning Objectives

1. Ingest data from an SPSS file, using the [haven](#) package.
2. Appropriately prepare data, check assumptions, execute appropriate modeling strategies, and interpret results

! Important

Your response to **every** Lab 6 task, whether we explicitly ask for it or not, should include a complete English sentence responding to the question. Code alone is not a sufficient response, even if the code is correct. Some responses might not need any code, but every response needs at least one complete sentence.

The Data for Lab 6

Lab 6 uses simulated data which reflect a clinical trial of a new drug to see its effect on a subject's systolic blood pressure (SBP). For our purposes, if a subject's SBP is over 130 mm Hg they are considered to have hypertension. In this trial, the goal was to reduce the SBP from a baseline level by using a new drug (Treatment C), and comparing that to the current top-of-the-line drug (Treatment B), and the oldest drug (Treatment A). The trial focused specifically on non-Hispanic African-American women who were in long-term relationships and between the ages of 55 and 65 years old, with a minimal comorbidity profile. The outcome of interest is the post-treatment systolic blood pressure ([sbp_follow](#)), and we are also given the subject's age, their pre-treatment systolic blood pressure ([sbp_baseline](#)) and whether or not the subject's partner has hypertension.

Variable	Description
subjectid	unique subject identifier
group	treatment group where, 1 = Group A, 2 = Group B, 3 = Group C

Variable	Description
<code>partner</code>	whether or not the subject's partner also has hypertension
<code>age</code>	subject age in years at baseline
<code>sbp_baseline</code>	subject's baseline systolic blood pressure (mm Hg)
<code>sbp_follow</code>	subject's follow-up systolic blood pressure (mm Hg)

The data has been made available to you in an SPSS file called `lab06.sav` available on [our 431-data page](#).

1 Task 1 (5 points)

Ingest the data, and then make sure that (a) group is a meaningfully leveled factor, (b) partnership status is a factor, and (c) age and the two blood pressure results are properly numeric variables, and to look for missing data. Once you've done this, run some simple and attractive, well-annotated summaries to report the number of subjects per group, as well as the partnership status, age and baseline blood pressure levels of the subjects in each group. Do the groups look comparable on these three baseline variables?

2 Task 2 (5 points)

Next, we'd like to know if our outcome, `sbp_follow`, seems appropriately modeled with a Normal distribution. Create a figure containing at least two nicely annotated panels (presented together using tools from the patchwork package) to assess whether a Normal model might be appropriate for this outcome. One of your panels should include a well-labeled Normal Q-Q plot. Then write a sentence of two to describe what you've done and your conclusions.

3 Task 3 (5 points)

Produce a figure to compare the three groups that allows you to assess the Normality and Equal Variances assumptions of an ANOVA to compare the SBP at follow-up means across the three treatment groups, ignoring all other information available in the data. What conclusions can you draw about ANOVA assumptions in this setting?

4 Task 4 (5 points)

Now complete the comparison of the SBP at follow-up means of the three treatment groups (A, B and C) using an analysis of variance. Call this `model4`. State the p value for treatment group that you obtain, and comment on the size of the differences in group means that you observe. Do not fall into the trap of assessing statistical significance, though. What fraction of the variation in SBP at follow-up is accounted for by `model4`?

5 Task 5 (5 points)

Augment your results in Task 4 by incorporating baseline SBP levels into the comparison. Call this `model5`. How does the p value for treatment group change in this revised model? What fraction of the variation in SBP at follow-up is accounted for by `model5`?

6 Task 6 (5 points)

Now, create `model6` by augmenting the model you fit in Task 5 to see if `partner` may also play a meaningful predictive role in a model for our outcome. Again, interpret whether the model's quality of fit has improved, and discuss what the addition of `partner` did to your estimates about the impact of the treatment groups on the outcome.

7 Task 7 (10 points)

Now, to create your final model (`model7`), instead of adjusting for `partner`, take into account the subject's `age` as well as the baseline systolic blood pressure.

Then build a comparison of the four models you've fit (in models `model4` through `model7`) in terms of the quality of fit (as measured by R-square, adjusted R-square, sigma, AIC and BIC) in the available data. What conclusions can you draw about fit quality for in comparing these four models? Does one model stand out as better or worse than the others? Why or why not?

8 Task 8 (10 points)

Throughout the above questions, we've been presented with a number of p values. In Chapter 10 of Spiegelhalter's *The Art of Statistics*, there is a robust discussion of p values. Write a short essay (150 words would be sufficient), which applies what you've learned from Spiegelhalter's Chapter 10 to the analyses you completed in Tasks 1-7.

9 Section 9 of your Lab Report: AI Usage

All students should include an AI Usage section in each assignment for this class. See the instructions from Lab 1 for more details.

10 Section 10 of your Lab Report: Session Information

Include the session information as a final section in this Lab. I've done so at the bottom of this document.

11 Additional Notes and Instructions

11.1 Submitting this Lab

Submit this Lab via [Canvas](#), using the Lab 6 assignment. Be sure to submit both:

1. Your Quarto file (.qmd), and
2. The HTML file you obtain by knitting the Quarto file (.html)

Be sure that your Quarto (and thus HTML) files include the **AI Usage** and **Session information** sections at the end of the document, and that these section headings appear in the Table of Contents.

11.2 Grading this Lab

This Lab will be graded by the TAs and then reviewed by Dr. Love. Your grades will be available one week after the Lab deadline.

The maximum score on this Lab is 50 points.

As each Lab passes its deadline (as listed in the [Course Calendar](#)), we will:

- post the answer sketch (48 hours after the deadline) and draft grading rubric to our Shared Google Drive, and then
- post grades and any revisions to the grading rubric or answer sketch one week after the deadline to a location we will provide to you.

11.3 Emergencies and Late Policy

We do not grant extensions on Lab deadlines.

- To receive full credit on a Lab, it must be received on Canvas no later than 59 minutes after the posted deadline. (This allows for small issues with uploading to Canvas to occur without penalty.)
 - Labs that are turned in 1-48 hours after the deadline will lose 10 points for late work.
- No extensions to Lab deadlines will be made this semester. Labs turned in more than 48 hours after the deadline will receive no credit, since by then the Lab Sketch will be posted.
- For details on how Lab grades affect your course grade, see the Syllabus.
- If you have an emergency that will keep you from submitting the Lab by even the late deadline of Friday at noon, please let Dr. Love know that (as soon as possible) via email and he will consider excusing you from the Lab.

11.4 Lab Regrade Requests

If, after your Lab is graded, you want Dr. Love to review the grading or correct a grading error, please follow the Lab Regrade Request policy [posted on our Labs page](#).

12 Session Information

```
xfun::session_info()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Locale:
  LC_COLLATE=English_United States.utf8
```

```
LC_CTYPE=English_United States.utf8
LC_MONETARY=English_United States.utf8
LC_NUMERIC=C
LC_TIME=English_United States.utf8
```

Package version:

askpass_1.2.1	backports_1.5.1	base64enc_0.1.6
bit_4.6.0	bit64_4.8.2	blob_1.3.0
broom_1.0.13	bslib_0.11.0	cachem_1.1.0
callr_3.8.0	cellranger_1.1.0	cli_3.6.6
clipr_0.8.1	compiler_4.6.1	conflicted_1.2.0
cpp11_0.5.5	crayon_1.5.3	curl_7.1.0
data.table_1.18.4	DBI_1.3.0	dbplyr_2.6.0
digest_0.6.39	dplyr_1.2.1	dtplyr_1.3.3
evaluate_1.0.5	farver_2.1.2	fastmap_1.2.0
fontawesome_0.5.3	forcats_1.0.1	fs_2.1.0
gargle_1.6.1	generics_0.1.4	ggplot2_4.0.3
glue_1.8.1	googledrive_2.1.2	googlesheets4_1.1.2
graphics_4.6.1	grDevices_4.6.1	grid_4.6.1
gtable_0.3.6	haven_2.5.5	highr_0.12
hms_1.1.4	htmltools_0.5.9	httr_1.4.8
ids_1.0.1	isoband_0.3.0	jquerylib_0.1.4
jsonlite_2.0.0	knitr_1.51	labeling_0.4.3
lifecycle_1.0.5	lubridate_1.9.5	magrittr_2.0.5
memoise_2.0.1	methods_4.6.1	mime_0.13
modelr_0.1.11	openssl_2.4.2	otel_0.2.0
pillar_1.11.1	pkgconfig_2.0.3	prettyunits_1.2.0
processx_3.9.0	progress_1.2.3	ps_1.9.3
purrr_1.2.2	R6_2.6.1	ragg_1.5.2
rappdirs_0.3.4	RColorBrewer_1.1-3	readr_2.2.0
readxl_1.5.0	rematch_2.0.0	rematch2_2.1.2
reprex_2.1.1	rlang_1.3.0	rmarkdown_2.31
rstudioapi_0.19.0	rvest_1.0.5	S7_0.2.2
sass_0.4.10	scales_1.4.0	selectr_0.6.0
stats_4.6.1	stringi_1.8.7	stringr_1.6.0
sys_3.4.3	systemfonts_1.3.2	textshaping_1.0.5
tibble_3.3.1	tidyr_1.3.2	tidyselect_1.2.1
tidyverse_2.0.0	timechange_0.4.0	tinytex_0.60
tools_4.6.1	tzdb_0.5.0	utf8_1.2.6
utils_4.6.1	uuid_1.2.2	vctrs_0.7.3
viridisLite_0.4.3	vroom_1.7.1	withr_3.0.3
xfun_0.60	xml2_1.6.0	yaml_2.3.12